



Part 1 - Data Collection & Preprocessing
CISB5123 (Text Analytics)
section 01AT

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1. Project Scope

Problem Statement

The project focuses on analyzing customer reviews of the Grab application to understand user sentiment and identify key issues affecting user satisfaction. With the increasing reliance on ride-hailing and delivery services, understanding customer feedback is essential for improving service quality and user experience.

Objectives

The main objectives of this project are:

- To classify user reviews into positive, negative, or neutral sentiments
- To identify common themes and issues mentioned in user feedback
- To provide insights that can help improve service performance and customer satisfaction

Target Audience

- Grab company stakeholders (product managers, developers, customer support teams)
- Researchers and analysts studying customer behavior
- General users interested in service quality trends

Ethical Considerations

- **Data Privacy:** Only publicly available reviews are used; no personal identifiable information is included
- **Bias:** Reviews may not represent all users equally (e.g., more negative reviews tend to be posted)
- **Consent:** Data is collected from publicly accessible sources where user consent is implied

2. Dataset Description

Data Source

The dataset consists of Grab user reviews collected from the `google-play-scraper` library to collect user reviews for the Grab app (`com.grabtaxi.passenger`). The data was stored in a CSV file format.

Dataset Size and Structure

- File: grab_reviews_preprocessed.csv
- Number of records: 1000
- Format: CSV

Data Limitations

- Reviews may be **biased toward extreme opinions** (very positive or very negative)

- Presence of **noise** such as slang, abbreviations, or mixed languages
- Some reviews may be **short or unclear**, reducing analytical value
- Possible imbalance in sentiment distribution

3. Text Preprocessing Techniques

Text preprocessing is essential to clean and transform raw text into a structured format suitable for analysis and modeling.

Steps Applied

1. Text Cleaning

- Removed special characters, punctuation, and numbers
- Converted all text to lowercase

Purpose:

Ensures consistency and reduces noise in the dataset

2. Tokenization

- Split text into individual words (tokens)

Purpose:

Allows the model to process text at the word level

3. Stop Word Removal

- Removed common words such as “the”, “is”, “and”

Purpose:

Eliminates words that do not contribute meaningful information

4. Stemming/Lemmatization

- Reduced words to their base or root form

Purpose:

Helps group similar words together and reduces dimensionality

5. Removal of Empty or Null Values

- Dropped rows with missing or invalid review text

Purpose:

Ensures data quality and prevents errors during analysis

Why These Steps Are Necessary

- Raw text data is **unstructured and inconsistent**
- Machine learning models require **clean and standardized input**
- Reduces noise and improves **accuracy and efficiency** of analysis

Challenges Faced

- Handling **slang and informal language** commonly found in reviews
- Dealing with **mixed languages** (e.g., English + Malay)
- Some reviews were **too short or meaningless** (e.g., “bad”, “ok”)
- Removing noise without losing important context

4. Conclusion

This project establishes a foundation for analyzing customer sentiment using text analytics. By defining a clear scope, preparing a structured dataset, and applying essential preprocessing techniques, the data is now ready for further steps such as sentiment classification and topic modeling.